

Zhidan Liu

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EMPLOYMENT

- 2024.08 - Now Assistant Professor, **The Hong Kong University of Science and Technology (Guangzhou)**, China
- 2022.07 - 2024.07 Associate Professor, **Shenzhen University**, China
- 2017.07 - 2022.06 Assistant Professor, **Shenzhen University**, China
- 2015.01 - 2017.06 Research Fellow, **Nanyang Technological University**, Singapore (Advisor: *Prof. Mo Li*)

EDUCATION

- 2009.09 - 2014.09 Ph.D. in Computer Science, **Zhejiang University**, Hangzhou, China, Advisor: *Prof. Dongming Lu*
- 2012.10 - 2013.11 Visiting student, **Nanyang Technological University**, Singapore, Advisor: *Prof. Mo Li*
- 2005.09 - 2009.07 B.E. in Computer Science, **Northeastern University**, Shenyang, China

RESEARCH INTERESTS

Artificial Internet of Things, Urban Computing, Mobile Computing, Data Mining and Analytic

HONORS AND AWARDS

- 2025.08 CCF Transactions on Pervasive Computing and Interaction *Best Reviewer Award*
- 2021.04 Outstanding Undergraduate Teacher Award (Shenzhen University)
- 2020.12 Excellent Class Director of Shenzhen University Tencent Mentor Award
- 2020.12 IEEE ICPADS'20 Best Paper Award
- 2018.12 Award of Tencent "Rhinceros Birds" - Scientific Research Foundation for Young Teachers
- 2018.06 Shenzhen Nanshan District Pilot Talent Award
- 2018.01 Shenzhen Peacock Plan Talent Award

PUBLICATIONS

([†] indicates the students I advised or co-advised, * indicates the corresponding author)

Refereed Journal Publications

1. [TITS] Jiyu Wang[†], **Zhidan Liu***. "A Two-Stage Imputation Method for Urban Traffic Data Based on Sparse Spatiotemporal Attention", *IEEE Transactions on Intelligent Transportation Systems*, Vol. xx, Issue xx, Pages xxxx-xxxx, March 2026. (IF: 8.400, [CCF-Rank B](#), JCR Q1)
2. [TMC] Zhenyu Yang[†], Qiang Yang, **Zhidan Liu***, Zhenjiang Li, Yongpan Zou, Kaishun Wu. "ArmPad: Transforming Forearms into Interaction Interfaces with Smartwatches", *IEEE Transactions on Mobile Computing*, Vol. xx, Issue xx, Pages xxxx-xxxx, March 2026. (IF: 9.200, [CCF-Rank A](#), JCR Q1)
3. [TCSS] Haodong Liu, Ye Zhang, **Zhidan Liu**, Hongyuan Zhu, Changhong Wang. "Heterogeneous Multi-Modal Multi-Label Behavioral Context Recognition Using Wearable Sensors and Mobile Devices", *IEEE Transactions on Computational Social Systems*, Vol. xx, Issue xx, Pages xxxx-xxxx, March 2026. (IF: 4.900, JCR Q1)
4. [TMC] Xiaosi Liu[†], Xiaowen Xu[†], **Zhidan Liu***, Zhenjiang Li, Kaishun Wu. "Spatio-Temporal Diffusion Model for Cellular Traffic Generation", *IEEE Transactions on Mobile Computing*, Vol. 25, Issue 1, Pages 257-271, January 2026. (IF: 9.200, [CCF-Rank A](#), JCR Q1)
5. [TITS] **Zhidan Liu**, Zhengze Sun[†], Junru Zhang, Bolin Zhang[†], Panrong Tong. "Regional Knowledge Transfer for Urban Traffic Flow Prediction via Satellite Imagery Assisted Contrastive Domain Adaptation", *IEEE Transactions on Intelligent Transportation Systems*, Vol. 26, Issue 11, Pages 19120-19134, November 2025. (IF: 8.400, [CCF-Rank B](#), JCR Q1)
6. [TMC] Zixiao Wang[†], Qi Dong, Tianzhang Xing*, **Zhidan Liu***, Zhenjiang Li, Xiaojiang Chen. "Enabling Effective OOD Detection via Plug-and-Play Network for Mobile Visual Applications", *IEEE Transactions on Mobile Computing*, Vol. 24, Issue 11, Pages 12471-12486, November 2025. (IF: 7.700, [CCF-Rank A](#), JCR Q1)
7. [TMC] Junru Zhang, Cheng Peng, **Zhidan Liu**, Lang Feng, Yuhan Wu, Yabo Dong, Duanqing Xu. "DI2SDiff++: Activity Style Decomposition and Diffusion-Based Fusion for Cross-Person Generalization in Activity Recognition", *IEEE Transactions on Mobile Computing*, Vol. 24, Issue 10, Pages 10760-10777, October 2025. (IF: 7.700, [CCF-Rank A](#), JCR Q1)

8. [TMC] **Zhidan Liu**, Guofeng Ouyang[†], Bolin Zhang[†], Bo Du, Chao Chen, Kaishun Wu. “Joint Order Dispatching and Vehicle Repositioning for Dynamic Ridesharing”, *IEEE Transactions on Mobile Computing*, Vol. 24, Issue 4, Pages 2628-2643, April 2025. (IF: 7.700, **CCF-Rank A**, JCR Q1)
9. [TKDE] **Zhidan Liu**, Yingqian Zhou[†], Xiaosi Liu[†], Haodi Zhang, Yabo Dong, Dongming Lu, Kaishun Wu. “Learning Road Network Index Structure for Efficient Map Matching”, *IEEE Transactions on Knowledge and Data Engineering*, Vol. 37, Issue 1, Pages 423-437, January 2025. (IF: 8.900, **CCF-Rank A**, JCR Q1)
10. [TBD] Biyao Gong[†], Tianzhang Xing*, **Zhidan Liu***, Wei Xi, Xiaojiang Chen. “Adaptive Client Clustering for Efficient Federated Learning over Non-IID and Imbalanced Data”, *IEEE Transactions on Big Data*, Vol. 10, Issue 6, Pages 1051-1065, December 2024. (IF: 3.344, JCR Q1)
11. [TMC] Xianjing Wu[†], **Zhidan Liu***, Zhenjiang Li, Shengjie Zhao*. “Greta: Towards A General Roadside Unit Deployment Framework”, *IEEE Transactions on Mobile Computing*, Vol. 23, Issue 7, Pages 7602-7617, July 2024. (IF: 7.900, **CCF-Rank A**, JCR Q1)
12. [TMC] Biyao Gong[†], Tianzhang Xing*, **Zhidan Liu***, Wei Xi, Xiaojiang Chen. “Towards Hierarchical Clustered Federated Learning with Model Stability on Mobile Devices”, *IEEE Transactions on Mobile Computing*, Vol. 23, Issue 6, Pages 7148-7164, June 2024. (IF: 7.900, **CCF-Rank A**, JCR Q1)
13. [TMC] **Zhidan Liu**, Hongquan Zhang[†], Guofeng Ouyang[†], Junyang Chen, Kaishun Wu. “Data-Driven Pick-Up Location Recommendation for Ride-Hailing Services”, *IEEE Transactions on Mobile Computing*, Vol. 23, Issue 2, Pages 1001-1015, February 2024. (IF: 7.900, **CCF-Rank A**, JCR Q1)
14. [TMC] **Zhidan Liu**, Jiancong Liu[†], Xiaowen Xu[†], Kaishun Wu. “DeepGPS: Deep Learning Enhanced GPS Positioning in Urban Canyons”, *IEEE Transactions on Mobile Computing*, Vol. 23, Issue 1, Pages 376-392, January 2024. (IF: 7.900, **CCF-Rank A**, JCR Q1)
15. [TMC] **Zhidan Liu**, Junhong Zheng[†], Jinye Lin[†], Liang Wang, Kaishun Wu. “Radar: Adversarial Driving Style Representation Learning with Data Augmentation”, *IEEE Transactions on Mobile Computing*, Vol. 22, No. 12, Pages 7070-7085, December 2023. (IF: 7.900, **CCF-Rank A**, JCR Q1)
16. [MONE] Biyao Gong[†], Tianzhang Xing*, **Zhidan Liu***, Junfeng Wang, Xiuya Liu. “Adaptive Clustered Federated Learning for Heterogeneous Data in Edge Computing”, *Mobile Networks and Applications*, DOI: <https://doi.org/10.1007/s11036-022-01978-8>, 2022. (IF: 3.426, JCR Q2)
17. [CJC] **Zhidan Liu**, Weixin Lin[†], Kaishun Wu. “A Spatial Structure Matching Algorithm for Large Spatial-Textual Datasets” (in Chinese), *Chinese Journal of Computers*, Vol. 45, Issue 6, Pages 1261-1275, June 2022. (**CCF-Rank A**)
18. [TIST] Liang Wang, Zhiwen Yu, Bin Guo, Dingqi Yang, Lianbo Ma, **Zhidan Liu**, Fei Xiong. “Data-Driven Targeted Advertising Recommendation System for Outdoor Billboard”, *ACM Transactions on Intelligent Systems and Technology*, Vol. 13, No. 2, Article 29, Pages 1-23, January 2022. (IF: 10.489, JCR Q1)
19. [IOTJ] **Zhidan Liu**, Zengyang Gong[†], Jiangzhou Li[†], Kaishun Wu, “mT-Share: A Mobility-Aware Dynamic Taxi Ridesharing System”, *IEEE Internet of Things Journal*, Vol. 9, Issue 1, Pages 182-198, January 2022. (IF: 10.238, JCR Q1)
20. [TITS] **Zhidan Liu**, Jiangzhou Li[†], Kaishun Wu. “Context-Aware Taxi Dispatching at City-Scale Using Deep Reinforcement Learning”, *IEEE Transactions on Intelligent Transportation Systems*, Vol. 23, Issue 3, Pages 1996-2009, March 2022. (IF: 9.551, **ESI Highly Cited Paper**, **CCF-Rank B**, JCR Q1)
21. [IOTJ] Shiwei Song, **Zhidan Liu**, Zhenjiang Li, Tianzhang Xing, Dingyi Fang. “Coverage-Oriented Task Assignment for Mobile Crowdsensing”, *IEEE Internet of Things Journal*, Vol. 7, Issue 8, Pages 7407-7418, April 2020. (IF: 9.936, JCR Q1)
22. [IOTJ] **Zhidan Liu**, Zhenjiang Li, Kaishun Wu. “UniTask: A Unified Task Assignment Design for Mobile Crowdsourcing based Urban Sensing”, *IEEE Internet of Things Journal*, Vol.6 Issue 4, Pages 6629-6641, August 2019. (IF: 9.936, JCR Q1)
23. [TMC] **Zhidan Liu**, Pengfei Zhou, Zhenjiang Li, Mo Li. “Think Like A Graph: Real-Time Traffic Estimation at City-Scale”, *IEEE Transactions on Mobile Computing*, Vol. 18, Issue 10, Pages 2446-2459, October 2019. (IF: 5.112, **CCF-Rank A**, JCR Q1)
24. [IEEENet] **Zhidan Liu**, Zhenjiang Li, Kaishun Wu, Mo Li. “Urban Traffic Prediction from Mobility Data Using Deep Learning”, *IEEE Network Magazine*, Vol. 32, Issue 4, Pages 40-46, August 2018. (IF: 8.808, JCR Q1)

25. [TITS] **Zhidan Liu**, Shiqi Jiang, Pengfei Zhou, and Mo Li. “A Participatory Urban Traffic Monitoring System: The Power of Bus Riders”, *IEEE Transactions on Intelligent Transportation Systems*, Vol. 18, Issue 10, Pages 2851-2864, October 2017. (IF: 4.051, [CCF-Rank B](#), JCR Q1)
26. [TON] **Zhidan Liu**, Zhenjiang Li, Mo Li, Wei Xing, and Dongming Lu. “Path Reconstruction in Dynamic Wireless Sensor Networks Using Compressive Sensing”, *IEEE/ACM Transactions on Networking*, Vol. 24, Issue 4, Pages 1948-1960, August 2016. (IF: 3.376, [CCF-Rank A](#), JCR Q1)
27. [TITS] **Zhidan Liu**, Zhenjiang Li, Mo Li, Wei Xing, and Dongming Lu. “Mining Road Network Correlation for Traffic Estimation via Compressive Sensing”, *IEEE Transactions on Intelligent Transportation Systems*, Vol. 17, Issue 7, Pages 1880-1893, July 2016. (IF: 3.724, [CCF-Rank B](#), JCR Q1)
28. [IJDSN] Bo Zeng, Yabo Dong, **Zhidan Liu**. “A Negotiation-Based TDMA Scheduling with Consecutive Slots Assignment for Wireless Sensor Networks”, *International Journal of Distributed Sensor Networks*, 2014. (IF: 1.239, JCR Q4)
29. [IJDSN] **Zhidan Liu**, Wei Xing, Yongchao Wang, Dongming Lu. “Hierarchical Spatial Clustering in Multihop Wireless Sensor Networks”, *International Journal of Distributed Sensor Networks*, 2013. (IF: 1.239, JCR Q4)

Refereed Conference Publications

1. [KDD’26] Fansheng Sun[†], Jiyu Wang[†], **Zhidan Liu***. “Orchestrating Reasoning and Reaction: An Asynchronous Hierarchical Framework for LLM-driven Traffic Signal Control”, In ACM KDD, Jeju, South Korea, August 2026. (Acceptance Rate: ~18.5%, [CCF-Rank A](#), [CSRanking Conference](#))
2. [ACL’26] Quyu Kong, Xu Zhang, Zhenyu Yang[†], Nolan Gao, Chen Liu, Panrong Tong, chenglin cai, Hanzhang Zhou, Jianan Zhang, Liangyu Chen, **Zhidan Liu**, Steven Hoi, Yue Wang. “MobileWorld: Benchmarking Autonomous Mobile Agents in Agent-User Interactive and MCP-Augmented Environments”, In *ACL (Main Conference)*, San Diego, California, United States, July 2026. (Acceptance Rate: 19%, [CCF-Rank A](#), [CSRanking Conference](#))
3. [ACL’26] Haodi Zhang, Xinrui Zhu, Mingze Kong, **Zhidan Liu**, Tao Fan, Kaishun Wu, Yuanfeng Song. “MTO: A Multi-turn Conversational Text-to-OverpassQL Dataset for Enhanced OpenStreetMap Query Generation”, In *ACL (Findings)*, San Diego, California, United States, July 2026. ([CSRanking Conference](#))
4. [GPC’25] Yaqi Huang[†], Zhengze Sun[†], **Zhidan Liu***. “Accelerating Graph Similarity Search with Maximum Common Subgraphs”, In *The 20th International Conference on Green, Pervasive, and Cloud Computing (GPC)*, Kuala Lumpur, Malaysia, December 2025.
5. [CIKM’25] Bolin Zhang[†], **Zhidan Liu***. “DSETA: Driving Style-Aware Estimated Time of Arrival”, In *ACM CIKM*, Seoul, Korea, November 2025. (Acceptance Rate: $443/1627 = 27\%$, [CCF-Rank B](#))
6. [ICDM’24] **Zhidan Liu**, Jinye Lin[†], Zhiyu Xia[†], Chao Chen, Kaishun Wu. “Towards Efficient Ridesharing via Order-Vehicle Pre-Matching Using Attention Mechanism”, In *IEEE ICDM*, Abu Dhabi, UAE, December 2024. (Acceptance Rate: $66/604 = 11\%$, [CCF-Rank B](#))
7. [KDD’24] Junru Zhang, Lang Feng, **Zhidan Liu**, Yuhan Wu, Yang He, Yabo Dong, Duanqing Xu. “Diverse Intra- and Inter-Domain Activity Style Fusion for Cross-Person Generalization in Activity Recognition”, In *ACM KDD*, Barcelona, Spain, August 2024. (Acceptance Rate: 20%, [CCF-Rank A](#))
8. [ICPADS’23] Meiquan Lai[†], Yaqi Huang[†], **Zhidan Liu***, Kaishun Wu. “An Optimized Lossless Graph Summarization for Large-Scale Graphs”, In *IEEE ICPADS*, Ocean Flower Island, Hainan, China, December 2023.
9. [MM’23] Mengzhu Wang, Junyang Chen, Huan Wang, Huisi Wu, **Zhidan Liu**, Qin Zhang. “Interpolation Normalization for Contrast Domain Generalization”, In *ACM MM*, Ottawa, Canada, October 2023. ([CCF-Rank A](#))
10. [ICPADS’22] Jianlong Xu, Zhiyu Xia[†], Yuhui Li, Yuxiang Zeng, **Zhidan Liu**. “Subgraph Sampling for Inductive Sparse Cloud Services QoS Prediction”, In *IEEE ICPADS*, Nanjing, China, January 2023.
11. [DASFAA’21] **Zhidan Liu**, Junhong Zheng[†], Zengyang Gong[†], Haodi Zhang, Kaishun Wu. “Exploiting Multi-Source Data for Adversarial Driving Style Representation Learning”, In *DASFAA*, Taipei, Taiwan, April 2021. ([CCF-Rank B](#))
12. [HPCC’20] Baofu Huang[†], **Zhidan Liu***, Kaishun Wu. “Structure Preserved Graph Reordering for Fast Graph Processing Without the Pain”, In *IEEE HPCC*, Fiji, December 2020. (Acceptance Rate: $58/379 = 15\%$)
13. [ICPADS’20] Baofu Huang[†], **Zhidan Liu***, Kaishun Wu. “Accelerating PageRank in Shared-Memory for Efficient Social Network Graph Analytics”, In *IEEE ICPADS*, HongKong, China, December 2020. ([Best Paper Award](#))

14. [ICCCN'20] Zengyang Gong[†], Bo Du, **Zhidan Liu**, Wei Zeng, Pascal Perez, Kaishun Wu. "SD-seq2seq: A Deep Learning Model for Bus Bunching Prediction Based on Smart Card Data", In *IEEE ICCCN*, Honolulu, Hawaii, USA, August 2020.
15. [ICDE'20] **Zhidan Liu**, Zengyang Gong[†], Jiangzhou Li[†], Kaishun Wu. "Mobility-Aware Dynamic Taxi Ridesharing", In *IEEE ICDE*, Dallas, Texas, USA, April 2020. (CCF-Rank A, CSRanking Conference)
16. [MSN'19] **Zhidan Liu**, Mingliang Huang[†], Zhi Ye[†], Kaishun Wu. "DeepRTP: A Deep Spatio-Temporal Residual Network for Regional Traffic Prediction", In *IEEE MSN*, Hong Kong, China, December 2019. (Invited paper)
17. [MobiSys'19] Yang Liu, Zhenjiang Li, **Zhidan Liu**, Kaishun Wu. "Real-time Arm Skeleton Tracking and Gesture Inference Tolerant to Missing Wearable Sensors", In *ACM MobiSys*, Seoul, Republic of Korea, June 2019. (Acceptance Rate: 40/174 = 22%, CCF-Rank B, CSRanking Conference)
18. [IPSN'18] Chu Cao, **Zhidan Liu**, Mo Li, Qin Zheng, Wenqiang Wang. "Walkway Discovery from Large Scale Crowdsensing", In *ACM/IEEE IPSN*, Porto, Portugal, April 2018. (Acceptance Rate: 22/82 = 26%, CCF-Rank B)
19. [MobiHoc'14] **Zhidan Liu**, Zhenjiang Li, Mo Li, Wei Xing, Dongming Lu. "Path Reconstruction in Dynamic Wireless Sensor Networks using Compressive Sensing", In *ACM MobiHoc*, Philadelphia, USA, August 2014. (Acceptance Rate: 40/211 = 18%, CCF-Rank B)
20. [ICCC'13] **Zhidan Liu**, Wei Xing, Yongchao Wang, Dongming Lu. "Hierarchical Spatial Clustering in Multi-hop Wireless Sensor Networks", In *IEEE ICC*, Xi'an, China, August 2013.
21. [AINA'13] **Zhidan Liu**, Wei Xing, Bo Zeng, Yongchao Wang, Dongming Lu. "Distributed Spatial Correlation-based Clustering for Approximate Data Collection in WSNs", In *IEEE AINA*, Barcelona, Spain, March 2013.
22. [ICACT'13] **Zhidan Liu**, Wei Xing, Yongchao Wang, Dongming Lu. "An Energy-Efficient Data Collection Scheme for Wireless Sensor Networks", In *IEEE ICACT*, PyeongChang, Korea, January 2013.
23. [CyberC'12] Bo Zeng, Yabo Dong, **Zhidan Liu**, Dongming Lu. "A Workload-Aware Link Scheduling for Heterogeneous Wireless Sensor Networks", In *IEEE CyberC*, Sanya, China, October 2012.
24. [WiCom'12] **Zhidan Liu**, Wei Xing, Yongchao Wang, Dongming Lu. "Adaptive Data Storage for Querying in Wireless Sensor Networks", In *IEEE WiCom*, Shanghai, China, September 2012.

Books, Poster, and Demo

1. [MobiCom'25] Xiaowen Xu, Xiaosi Liu, **Zhidan Liu***, Zhenjiang Li, Kaishun Wu. "Poster: Diffusion-Driven Spatio-Temporal Modeling of Cellular Traffic Generation", In *ACM MobiCom*, Hong Kong, China, November 2025.
2. [Book'21] **Zhidan Liu**, Kaishun Wu. "Mobility Data-Driven Urban Traffic Monitoring", *Springer*, Singapore, May, 2021.
3. [MobiSys'19] Yang Liu, Chengdong Lin, Zhenjiang Li, **Zhidan Liu**, Kaishun Wu. "Poster: When Wearable Sensing Meets Arm Tracking", In *ACM MobiSys*, Seoul, Republic of Korea, June 2019.
4. [IPSN'18] Chu Cao, **Zhidan Liu**, Mo Li, Qin Zheng, Wenqiang Wang. "Demo Abstract: Walkway Discovery from Large Scale Crowdsensing", In *ACM/IEEE IPSN*, Porto, Portugal, April 2018.

Patents

1. **Zhidan Liu**, Yingqian Zhou, Kaishun Wu. *A Method for Adaptively Adjusting the Search Radius in Map Matching Based on Reinforcement Learning* (ZL202310321723.0), 2026.02.06, Chinese invention patent
2. **Zhidan Liu**, Jiayue Ruan, Kaishun Wu. *Method and System for Adversarial Domain Adaptation-Based Human Activity Recognition Using IMU Data* (ZL202310336403.2), 2025.10.31, Chinese invention patent
3. **Zhidan Liu**, Yingqian Zhou, Kaishun Wu. *A Method for Constructing Road Network Spatial Index Based on Learned Index* (ZL202310321722.6), 2025.10.03, Chinese invention patent
4. **Zhidan Liu**, Yaqi Huang. *Graph Similarity Search Method, Device, Computer-Readable Storage Medium, and Program* (ZL202410540182.5), 2025.09.12, Chinese invention patent
5. **Zhidan Liu**, Weixin Lin, Kaishun Wu. *A Spatial Structure Matching Method for Large Spatial-Textual Datasets* (ZL202111190044.1), 2024.12.31, Chinese invention patent

6. **Zhidan Liu**, Jiancong Liu, Kaishun Wu. *A Deep Learning-Based GPS Enhancement Positioning Method and System for Urban Canyons* (ZL202210425764.X), 2024.10.29, Chinese invention patent
7. **Zhidan Liu**, Baofu Huang, Kaishun Wu. *A Lightweight and Efficient Method for Vertex Reordering in Graphs* (ZL202011134445.0), 2023.12.12, Chinese invention patent
8. **Zhidan Liu**, Baofu Huang, Mingliang Huang, Kaishun Wu. *Parallel Computing Acceleration Method for PageRank on Social Network Graph Data* (ZL201910999376.0), 2023.10.13, Chinese invention patent
9. **Zhidan Liu**, Junhong Zheng, Kaishun Wu. *A Method for Learning Driving Style Representation based on Multi-context Data* (ZL202011431508.9), 2023.01.13, Chinese invention patent
10. **Zhidan Liu**, Kaishun Wu. *A Unified Mobile Crowdsourcing Task Allocation Method for Urban Sensing* (ZL201910121165.7), 2022.09.20, Chinese invention patent
11. **Zhidan Liu**, Zenyang Gong, Jiangzhou Li, Kaishun Wu. *A Mobility-Aware Dynamic Ridesharing System* (ZL201910515242.7), 2022.09.20, Chinese invention patent
12. **Zhidan Liu**, Pengfei Zhou, Zhenjaing Li, Mo Li, Kaishun Wu. *A Traffic Estimation Method for Urban Road Networks* (ZL201811140012.9), 2021.09.24, Chinese invention patent
13. **Zhidan Liu**, Jiangzhou Li, Kaishun Wu. *A Deep Reinforcement Learning based Taxi Dispatching Method* (ZL202010521882.1), 2021.07.13, Chinese invention patent

SELECTED PROJECTS/GRANTS

- [PI] **Dynamic Optimization and Cooperative Decision-Making Mechanisms for Crowd-Powered Shared Mobility**, **National Natural Science Foundation of China (General Program)**, No.62572416, 2026.01 - 2029.12, 500K CNY. (Ongoing)
- [PI] **Research on the Key Technologies of Mobility Context Aware Personalized Ridesharing**, **National Natural Science Foundation of China (General Program)**, No.62172284, 2022.01 - 2025.12, 752K CNY. (Completed)
- [PI] **Research on Exploiting Big Mobility Trajectory Data for Smart Sensing of Urban Traffic**, **National Natural Science Foundation of China (Young Scientists Fund)**, No.61802261, 2019.01 - 2021.12, 300K CNY. (Completed)
- [PI] **Research on the Key Technologies of Shared Memory based Social Network Graph Computing**, **National Natural Science Foundation of Guangdong (General Program)**, No.2022A1515010155, 2022.01 - 2024.12, 100K CNY. (Completed)
- [PI] **Joint Research of Crowdsensing and Data Analytic for Smart City**, **National Natural Science Foundation of Guangdong (General Program)**, No.2020A1515011502, 2019.10 - 2022.09, 100K CNY. (Completed)
- [PI] **Research on Crowdsensing based Urban Sensing Framework**, **Natural Science Foundation of Shenzhen University Grant**, No.2018061, 2018.01 - 2022.12, 200K CNY. (Completed)
- [PI] **Research on Big Mobility Trajectory Data based Intelligent Transport Systems**, **Startup Fund for Shenzhen Overseas High-Caliber Personnel**, No.827/000344, 2018.01 - 2021.12, 2700K CNY. (Completed)
- [PI] **Research on Exploiting Big Trajectory Data for Intelligent Transport Sensing and Computing**, **Tencent "Rhinoceros Birds" - Scientific Research Foundation for Young Teachers of Shenzhen University**, No.201802, 2019.01 - 2020.12, 50K CNY. (Completed)
- [Co-I] **Key Innovation Team on Meteorological Service Support Technologies for Low-Altitude Flight**, **Science and Technology Innovation Team of Guangdong Provincial Meteorological Bureau**, GRMCTD02505-ZD05, 2025.09 - 2028.09, 67.5K CNY. (Ongoing)
- [Co-I] **Multi-dimensional and Deep Perception Information based Social Computing**, **Joint Funds of National Natural Science Foundation of China (Key Program)**, No.U2001207, 2021.01 - 2024.12, 2550K CNY. (Completed)

Membership:

- **IEEE Senior Member** (since 2024)
- **CCF Senior Member** (since 2023)
- Member of ACM (since 2018)
- Member of ACM SIGMOBILE (since 2021)
- Executive member of CCF Ubiquitous Computing Technical Committee (since 2022)
- Executive member of CCF Internet of Things Technical Committee (since 2023)
- Executive member of CCF Intelligent Transportation Division (since 2023)
- Executive member of ACM SIGSpatial China (since 2023)

Editorial Board:

- Young Editorial Board Member of **Journal of Social Computing**, 2024.02 - Now
- Lead Guest Editor: *Special Issue on Ubiquitous Sensing and Computing for Intelligent Transportation* in **Data Science for Transportation**

Conference Organization Chair:

- Special Session Chair: IEEE CSCWD 2022
- Local Chair: EAI ICECI 2021
- Conference Finance Chair: IEEE ICPADS 2017
- Conference Finance Chair: IEEE ICPADS 2018

Technical Program Committee Member:

- 2026: ACM KDD, IEEE ICDE (demo track), VLDB (demo track), The 20th International Conference on Green, Pervasive, and Cloud Computing (GPC), The 18th International Conference on Networking, Architecture, and Storage (NAS)
- 2025: ACM CSCW (**Associate Chair**), ACM KDD (**Outstanding Reviewer** for August Cycle), IEEE MSN, Australasian Database Conference (ADC)
- 2024: ACM KDD, IEEE ICDE (demo track), IEEE MSN
- 2023: ACM CIKM, IEEE ICPADS
- 2022: ACM KDD, ACM WSDM, IEEE ICPADS
- 2021: IEEE ICDCS, IEEE MASS
- 2020: IEEE ICDCS, IEEE HiPC
- 2019: IEEE MASS, IEEE MSN
- 2018: IEEE ICPADS, IEEE ICCCN
- 2017: IEEE ICPADS

Transaction and Journal Reviewer (selected):

- IEEE Transactions on Knowledge and Data Engineering (TKDE)
- IEEE Transactions on Mobile Computing (TMC)
- IEEE Transactions on Intelligent Transportation Systems (TITS)
- IEEE/ACM Transactions on Networking (TON)
- ACM Transactions on Spatial Algorithms and Systems (TSAS)
- IEEE Transactions on Wireless Communications (TWC)
- IEEE Transactions on Parallel and Distributed Systems (TPDS)
- ACM Transactions on Sensor Networks (TOSN)
- IEEE Transactions on Communications (TOC)
- IEEE Internet of Things Journal (JIOT)
- IEEE Communications Magazine
- IEEE Computational Intelligence Magazine
- Elsevier Pervasive and Mobile Computing (PMC)
- Ad Hoc & Sensor Wireless Networks (AHSWN)
- Elsevier Computer Networks

INVITED TALKS

- “*AI-Driven Dynamic Ridesharing*”, HMCC 2025 & HHME 2025, Dalian, China, August 2025
- “*AI-Driven Dynamic Ridesharing*”, UGOD 2025 Summer AI Symposium, HKUST (GZ), Guangzhou, China, July 2025
- “*The New Era of Artificial Intelligence: Opportunities and Challenges*”, Guangzhou Foreign Language School, Guangzhou, February 2025
- “*Data-Driven Pick-Up Location Recommendation for Ride-Hailing Services*”, CCF HHME 2024, Taiyuan, China, August 2024
- “*Deep Learning Enhanced GPS Positioning in Urban Canyons*”, Embedded AI Summit 2024, Dunhuang, China, August 2024
- “*Trajectory Data Driven Smart Mobility: Sensing, Computing, and Applications*”, HKUST (GZ), Guangzhou, China, September 2023
- “*Trajectory Data Driven Smart Transportation Research: Sensing and Computing*”, Zhejiang University, Hangzhou, China, April 2023
- “*Trajectory Data Driven Smart Transportation Research: Sensing and Computing*”, online, CCF HHME 2022, September 2022

TEACHING

Teaching Courses (at HKUST (Guangzhou)):

- (UG) C++ Programming (**UFUG2601**) Fall 2025
- (PG) Mobile Sensing and Computing for Transportation (**INTR6000N**) Spring 2025/2026

Teaching Courses (at Shenzhen University):

- (UG) Intelligent Networking and Computing Spring 2023
- (UG) Introduction to Wireless Sensor Networks Spring 2017-2022
- (UG) Internet Programming (in JAVA) Fall 2017-2023
- (PG) Guidance on Academic Writing Fall 2021-2023

Updated in May 2026